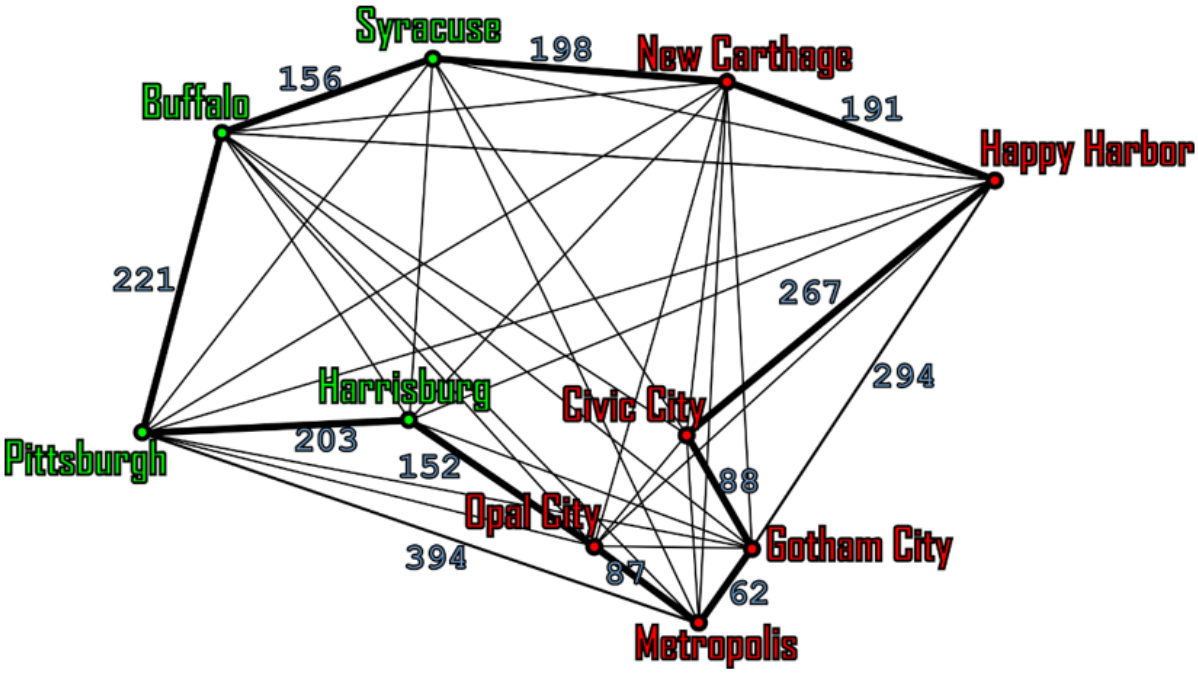


Figure 17.8 An example map, mixing DC universe's cities and other more real ones, with some distances: because this is a complete graph with close to fifty edges, we are only showing a few numbers, almost only for the edges in the best tour (the thicker edges).

The abstract formulation of this riddle is a well-known computer science problem known as the `traveling salesman problem` (*TSP*): it's, as you might have guessed, a difficult puzzle to solve, in particular a *NP-complete* problem; this implies it's both *NP-Hard* and in *NP* – the former meaning that there is no known deterministic algorithm that can solve it in polynomial time, while the latter means that, nevertheless, if we have a candidate solution, we can check it in polynomial time.

Informally, we can say that we expect any deterministic algorithm that solves TSP will need exponential time